MathCity.org Merging man and maths	**Some Important Derivative** **For BSc or BS Level** Available online @ http://www.mathcity.org, Version: 2.0

- $\frac{d}{dx}(c)=0$, where c is constant
- $\frac{d}{dx}(x^n)=nx^{n-1}$
- $\frac{d}{dx}[f(x)]^n=n[f(x)]^{n-1}\frac{d}{dx}[f(x)]$
- $\frac{d}{dx}\sin x=\cos x$
- $\frac{d}{dx}\cos x=-\sin x$
- $\frac{d}{dx}\tan x=\sec^2 x$
- $\frac{d}{dx}\cot x=-\csc^2 x$
- $\frac{d}{dx}\csc x=-\csc x\cot x$
- $\frac{d}{dx}\sec x=\sec x\tan x$
- $\frac{d}{dx}Sin^{-1}x=\frac{1}{\sqrt{1-x^2}}$
- $\frac{d}{dx}Cos^{-1}x=\frac{-1}{\sqrt{1-x^2}}$
- $\frac{d}{dx}Tan^{-1}x=\frac{1}{1+x^2}$
- $\frac{d}{dx}Cot^{-1}x=\frac{-1}{1+x^2}$
- $\frac{d}{dx}Sec^{-1}x=\frac{1}{x\sqrt{x^2-1}}$
- $\frac{d}{dx}Csc^{-1}x=\frac{-1}{x\sqrt{x^2-1}}$
- $\frac{d}{dx}a^x=a^x\ln a$
- $\frac{d}{dx}e^x=e^x$
- $\frac{d}{dx}\log_a x=\frac{1}{x\ln a}$
- $\frac{d}{dx}\ln x=\frac{1}{x}$
- $\frac{d}{dx}\sinh x=\cosh x$
- $\frac{d}{dx}\cosh x=\sinh x$
- $\frac{d}{dx}\tanh x=\operatorname{sech}^2 x$
- $\frac{d}{dx}\coth x=-\operatorname{csch}^2 x$
- $\frac{d}{dx}\operatorname{sech} x=-\operatorname{sech} x\tanh x$
- $\frac{d}{dx}\operatorname{csch} x=-\operatorname{csch} x\coth x$
- $\frac{d}{dx}Sinh^{-1}x=\frac{1}{\sqrt{x^2+1}}$
- $\frac{d}{dx}Cosh^{-1}x=\frac{1}{\sqrt{x^2-1}}$
- $\frac{d}{dx}Tanh^{-1}x=\frac{1}{1-x^2}$
- $\frac{d}{dx}Coth^{-1}x=\frac{1}{1-x^2}$
- $\frac{d}{dx}Sech^{-1}x=\frac{-1}{x\sqrt{1-x^2}}$
- $\frac{d}{dx}Csch^{-1}x=\frac{-1}{x\sqrt{1+x^2}}$

Some Standard nth Derivative

- $\frac{d^n}{dx^n}(ax+b)^m=\frac{m!}{(m-n)!}a^n(ax+b)^{m-n}$ if $m\geq n$
- $\frac{d^n}{dx^n}\left(\frac{1}{ax+b}\right)=\frac{(-1)^n n!a^n}{(ax+b)^{n+1}}$
- $\frac{d^n}{dx^n}[\ln(ax+b)]=\frac{(-1)^{n-1}(n-1)!a^n}{(ax+b)^n}$
- $\frac{d^n}{dx^n}e^{ax}=a^n e^{ax}$
- $\frac{d^n}{dx^n}\sin(ax+b)=a^n\sin\left(ax+b+n.\frac{\pi}{2}\right)$
- $\frac{d^n}{dx^n}\cos(ax+b)=a^n\cos\left(ax+b+n.\frac{\pi}{2}\right)$
- $\frac{d^n}{dx^n}e^{ax}\sin(bx+c)=(a^2+b^2)^{\frac{n}{2}}e^{ax}\sin\left(bx+c+n\tan^{-1}\frac{b}{a}\right)$
- $\frac{d^n}{dx^n}e^{ax}\cos(bx+c)=(a^2+b^2)^{\frac{n}{2}}e^{ax}\cos\left(bx+c+n\tan^{-1}\frac{b}{a}\right)$

Leibniz's Theorem

- $\frac{d^n}{dx^n}(uv)={}^nC_0u^{(n)}v+{}^nC_1u^{(n-1)}v'+{}^nC_2u^{(n-2)}v''+\ldots+{}^nC_{n-1}u'v^{n-1}+{}^nC_nuv^n$

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Differentiation of Trigonometric Functions

1) $\frac{d}{dx}(Sinx) = Cosx \frac{d}{dx}(x)$

2) $\frac{d}{dx}(Cosx) = -Sinx \frac{d}{dx}(x)$

3) $\frac{d}{dx}(tanx) = Sec^2x \frac{d}{dx}(x)$

4) $\frac{d}{dx}(Cotx) = -Cosec^2x \frac{d}{dx}(x)$

5) $\frac{d}{dx}(Secx) = Secx.tanx \frac{d}{dx}(x)$

6) $\frac{d}{dx}(Cesecx) = -Cosecx.Cotx \frac{d}{dx}(x)$

Differentiation of Inverse Trigonometric Functions

1) $\frac{d}{dx}(Sin^{-1}x) = \frac{1}{\sqrt{1-x^2}} \frac{d}{dx}(x)$

2) $\frac{d}{dx}(Cos^{-1}x) = \frac{-1}{\sqrt{1-x^2}} \frac{d}{dx}(x)$

3) $\frac{d}{dx}(tan^{-1}x) = \frac{1}{1+x^2} \frac{d}{dx}(x)$

4) $\frac{d}{dx}(Cot^{-1}x) = \frac{-1}{1+x^2} \frac{d}{dx}(x)$

5) $\frac{d}{dx}(Sec^{-1}x) = \frac{1}{|x|\sqrt{x^2-1}} \frac{d}{dx}(x)$

6) $\frac{d}{dx}(Cosec^{-1}x) = \frac{-1}{|x|\sqrt{x^2-1}} \frac{d}{dx}(x)$

Differentiation of Hyperbolic Functions

1) $\frac{d}{dx}(Sinhx) = Coshx \frac{d}{dx}(x)$

2) $\frac{d}{dx}(Coshx) = Sinhx \frac{d}{dx}(x)$

3) $\frac{d}{dx}(tanhx) = Sech^2x \frac{d}{dx}(x)$

4) $\frac{d}{dx}(Cothx) = -Cosech^2x \frac{d}{dx}(x)$

5) $\frac{d}{dx}(Sechx) = -Sechx.tanhx \frac{d}{dx}(x)$

6) $\frac{d}{dx}(Cosechx) = -Cosechx.cothx \frac{d}{dx}(x)$

Differentiation of Inverse Hyperbolic Functions

1) $\frac{d}{dx}(Sinh^{-1}x) = \frac{1}{\sqrt{1+x^2}} \frac{d}{dx}(x)$

2) $\frac{d}{dx}(Cosh^{-1}x) = \frac{1}{\sqrt{x^2-1}} \frac{d}{dx}(x)$

3) $\frac{d}{dx}(tanh^{-1}x) = \frac{1}{1-x^2} \frac{d}{dx}(x)$

4) $\frac{d}{dx}(Coth^{-1}x) = \frac{-1}{x^2-1} \frac{d}{dx}(x)$

5) $\frac{d}{dx}(Sech^{-1}x) = \frac{-1}{|x|\sqrt{1-x^2}} \frac{d}{dx}(x)$

6) $\frac{d}{dx}(Cosech^{-1}x) = \frac{-1}{|x|\sqrt{x^2+1}} \frac{d}{dx}(x)$

Integration Formulas

1 $\int \sin(ax+b)\, dx = -\frac{1}{a}\cos(ax+b) + C$

2 $\int \cos(ax+b)\, dx = \frac{1}{a}\sin(ax+b) + C$

3 $\int \tan(ax+b)\, dx = -\frac{1}{a}\ln\left|\cos(ax+b)\right| + C = \frac{1}{a}\ln\left|\sec(ax+b)\right| + C$

4 $\int \cot(ax+b)\, dx = \frac{1}{a}\ln\left|\sin(ax+b)\right| + C$

5 $\int \operatorname{cosec} x\, dx = \ln\left|\operatorname{cosec} x - \cot x\right| + C = ln\left|\tan\frac{x}{2}\right| + C$

6 $\int \sec x\, dx = \ln\left|\sec x + \tan x\right| + C = ln\left|\tan\left(\frac{\pi}{4}+\frac{x}{2}\right)\right| + C$

7 $\int e^{ax+b}\, dx = \frac{1}{a} \times e^{ax+b} + C$

8 $\int a^{bx+c}\, dx = \frac{1}{b\ln a} \times a^{bx+c} + C$

9 $\int \sqrt{x^2+a^2}\, dx = \frac{x}{2}\sqrt{x^2+a^2} + \frac{a^2}{2}\sinh^{-1}\left(\frac{x}{a}\right)$

10 $\int \sqrt{x^2-a^2}\, dx = \frac{x}{2}\sqrt{x^2-a^2} - \frac{a^2}{2}\cosh^{-1}\left(\frac{x}{a}\right)$

11 $\int \sqrt{a^2-\boldsymbol{x^2}}\, dx = \frac{x}{2}\sqrt{a^2-\boldsymbol{x^2}} + \frac{a^2}{2}\sin^{-1}\left(\frac{x}{a}\right)$

12 $\int \frac{1}{\sqrt{x^2+a^2}}\, dx = \sinh^{-1}\left(\frac{x}{a}\right) + C = ln\left(\frac{x+\sqrt{x^2+a^2}}{a}\right) + C$

13 $\int \frac{1}{\sqrt{x^2-a^2}}\, dx = \cosh^{-1}\left(\frac{x}{a}\right) + C = ln\left(\frac{x+\sqrt{x^2-a^2}}{a}\right) + C$

14 $\int \frac{1}{x^2+a^2}\, dx = \frac{1}{a}\tan^{-1}\left(\frac{x}{a}\right) + C$

15 $\int \frac{1}{\sqrt{a^2-x^2}}\, dx = \sin^{-1}\left(\frac{x}{a}\right) + C$

16 $\int \frac{1}{a^2-x^2}\, dx = \frac{1}{2a}\, ln\left|\frac{a+x}{a-x}\right| + C$

17 $\int \frac{1}{x^2-a^2}\, dx = \frac{1}{2a}\, ln\left|\frac{x-a}{x+a}\right| + C$

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